

*How different is targeted training data from random? And if it is different, is that difference attributable to the uncertainty signal – or to something the signal is merely correlated with?*

## SETUP



**held fixed in every arm:** the allocation *shape* (same softmax, same budget, same ranking, same greedy rule). Only what the rule is pointed *at* changes.

each control destroys one thing: SHUFFLED the signal's cluster assignment · CENTROID the signal entirely · DIRECTION the fitted axis · NULL everything

## MEASUREMENT

held-out Cohen's  $d$ , mean over 20 cells:

|                            |        |
|----------------------------|--------|
| TARGETED (the real signal) | +0.675 |
| SHUFFLED signal destroyed  | +0.668 |
| CENTROID arbitrary cluster | +0.740 |
| DIRECTION random axis      | +0.627 |
| NULL random vs random      | -0.133 |

## OUTCOME

paired increment, targeted *minus* control, over all 20 cells:  
 vs its own SHUFFLED control +0.0073 of a  $d$ , targeted wins 13/20 — a coin flip  
 vs an arbitrary CENTROID —0.0649, targeted wins 4/20 — it is *worse*

**destroying the signal reproduces the separation**  
 what targeting buys is **concentration**, not targeting